

Introduction to the Queueing Systems

The formation of queues is a frequent phenomenon that occurs whenever service demands exceed the permitted capacity of service devices. Queueing phenomena occur in the most diverse fields of activity: telephone stations, repair shops, ticket offices, the execution of tasks in a computer center, the organization of transport,....

Introduction to the Queueing Systems

The general model of a queueing phenomenon (waiting system) can be summarized as follows: requests (customers) arrive at a certain location and demand a certain service. If a service device is free, the incoming request immediately goes to this device, where it is served. Otherwise, there are two possibilities: either the request leaves the system (denied request systems), or it goes into a queue (queueing system). At some point, the request is selected for service according to a discipline. On completion of the service, the request leaves the system.

A queueing system therefore comprises a service area with one or more service devices (servers), and a waiting area in which a possible queue is formed.

Introduction to the Queueing Systems

To identify a queueing system, we need to specify the entry rate, the service mechanism and the waiting discipline.

- The size of a potential customer population (or the number of sources) can be finite or infinite.
- The process (flow) of arrivals can be regular or random. The time between two successive arrivals follows a probability law.
- The service mechanism includes the number of servers and the distribution service times.
- Queueing discipline or service discipline, customers can be selected and served in order of arrival **FCFS**, or **LCFS**, or selected at random **RANDOM**.
- Waiting space capacity can be unlimited or not.

Introduction to the Queueing Systems

Classification of queueing systems

We use a symbolic notation "**Kendall notation**" comprising 4 symbols arranged in the order $A/B/s/m$, where

- A and B describe respectively the distribution of times between two successive arrivals and the distribution of service times,
- s is the number of servers (connected in parallel),
- m is the system capacity (the number of servers plus the number of waiting positions).
- The last symbol can be omitted if $m = \infty$.

Introduction to the Queueing Systems

Classification of queueing systems

To specify distributions A and B , we introduce the following symbols:

M : Exponential distribution ;

E_k : Erlang distribution of degree k ;

H_k : Hyperexponential distribution of degree k ;

D : Deterministic ;

G : General distribution.

Markovian Queueing Systems

Mathematical analysis

The mathematical study of a queueing system involves the introduction of an appropriately defined stochastic process.

The stationary distribution of the introduced stochastic process allows us to obtain the system's **performance characteristics**, such as:

- the mean number of customers in system $\bar{n}$;
- the mean number of customers in the queue $\bar{n}_q$;
- the mean waiting time for a customer $\bar{W}$;
- the mean amount of time a customer spends in the system $\bar{W}_s$.

Markovian Queueing Systems

Mathematical analysis

One of a more important and useful relationship in queueing theory is what is commonly known as **Little's Law**:

$$\bar{n} = \lambda \bar{W}_s, \quad \bar{n}_q = \lambda \bar{W},$$

and

$$\bar{n} = \bar{n}_q + \frac{\lambda}{\mu}, \quad \bar{W}_s = \bar{W} + \frac{1}{\mu},$$

where λ is the rate at which customers enter the system, $\frac{1}{\mu}$ is the mean service time ($\mu > 0$).

Another important measure for a queueing system, the one that measures the degree of saturation of the system, called the traffic intensity ρ . It is defined by:

$$\rho = \frac{\text{mean service time}}{\text{mean time between two successive arrivals}}.$$

Queueing Model M/M/1

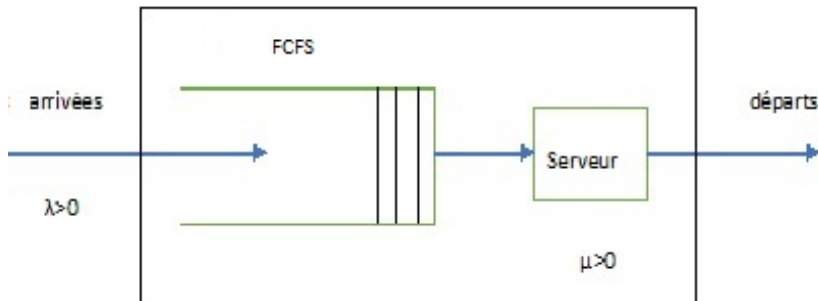
Model Description

Customers arrive at the system according to a Poisson process of rate $\lambda > 0$, (mean number of customers arriving during a unit of time) ; i.e., the time interval between two successive arrivals follows an exponential law with parameter $\lambda > 0$. The service is provided by a single server. When a customer arrives, if the server is free, the customer is immediately served. Otherwise, if the server is busy the customer joins the queue. The queue has unlimited capacity. Waiting discipline is FCFS. Service times follow an exponential distribution with parameter $\mu > 0$. Consequently, the service rate is μ (mean number of customers served during a unit of time), and the mean time of service for a customer is $\frac{1}{\mu}$.

Queueing Model M/M/1

Model Description

Finally, we assume that the r.v representing the times between two consecutive arrivals and the service times are mutually independent.



Queueing Model M/M/1

Model analysis

The state of the system at date t can be described by the stochastic process

$$\{N(t), t \geq 0\}, \quad (5)$$

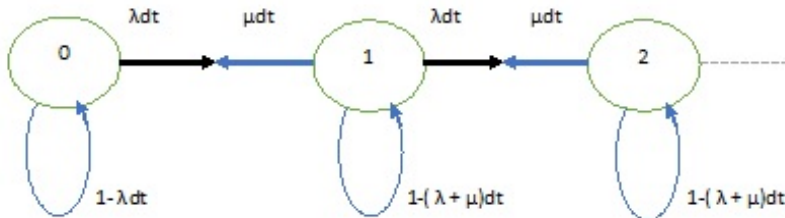
which is a birth and death process and states space

$S = \{0, 1, 2, \dots\}$. The rates of transitions are $\lambda_n = \lambda, \forall n \geq 0$ and $\mu_n = \mu, \text{ for } n \geq 1, (\mu_0 = 0)$.

Queueing Model M/M/1

Transitory State of the system

Let $p_n(t) = \mathbb{P}(N(t) = n), \forall n \geq 0$ be the probability of observing n customers in the system at time t . The representative transition graph of an M/M/1 queue is as follows



Queueing Model M/M/1

Transitory State of the system

From the representative graph, we obtain

$$\left\{ \begin{array}{l} p_0(t + \Delta t) = (1 - \lambda \Delta t) p_0(t) + \mu \Delta t p_1(t) \\ \text{-----} \\ p_n(t + \Delta t) = \lambda \Delta t p_{n-1}(t) + [1 - (\lambda + \mu) \Delta t] p_n(t) \\ \quad + \mu \Delta t p_{n+1}(t), n \geq 1. \end{array} \right.$$

Then, letting $\Delta t \rightarrow 0$, we get the **Kolmogorov equations**:

$$\left\{ \begin{array}{l} p'_0(t) = -\lambda p_0(t) + \mu p_1(t) \\ \text{-----} \\ p'_n(t) = \lambda p_{n-1}(t) - (\lambda + \mu) p_n(t) + \mu p_{n+1}(t), n \geq 1. \end{array} \right. \quad (6)$$

Queueing Model M/M/1

Stationary state of the system

Let $\lim_{t \rightarrow \infty} p_n(t) = p_n, n \geq 0$, exist and are independent of the initial state of the process. The balance equations are:

$$\left\{ \begin{array}{l} 0 = -\lambda p_0 + \mu p_1 \\ 0 = -(\lambda + \mu)p_1 + \lambda p_0 + \mu p_2 \\ \text{-----} \\ 0 = -(\lambda + \mu)p_n + \lambda p_{n-1} + \mu p_{n+1}, n \geq 1. \end{array} \right. \quad (7)$$

Hence, we get

$$p_n = \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^n = (1 - \rho) \rho^n, n \geq 0,$$

where $\rho = \frac{\lambda}{\mu} < 1$.

Queueing Model M/M/1

Performance Measures

- Mean number of customers in the system

$$\bar{n} = \frac{\rho}{(1-\rho)} = \frac{\lambda}{\mu-\lambda}.$$

- Mean number of customers in the queue

$$\bar{n}_q = \frac{\rho^2}{(1-\rho)} = \frac{\lambda^2}{\mu(\mu-\lambda)}.$$

- Mean time a customer spends in the system

$$\bar{W}_s = \frac{1}{\mu(1-\rho)} = \frac{1}{\mu-\lambda}.$$

- Mean waiting time for a customer $\bar{W} = \frac{\rho}{\mu(1-\rho)} = \frac{\lambda}{\mu(\mu-\lambda)}.$

Queueing Model M/M/1

Customer Waiting Time

The distribution of W the waiting time and W_s the sojourn time of a customer in an $M/M/1$ queueing system are given as follows:

$$P(0 \leq W \leq t) = 1 - \rho e^{-\mu(1-\rho)t}.$$

And

$$P(0 < W_s \leq t) = 1 - e^{-\mu(1-\rho)t}.$$

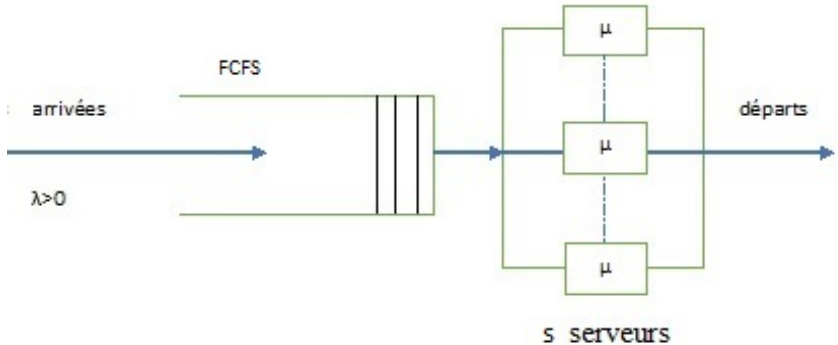
Queueing Model M/M/s

Model description

Customers arrive at the system according to a Poisson process of rate $\lambda > 0$. The service is provided by $s \geq 1$ servers connected in parallel. Upon the arrival of a customer, if one of the servers is free, the customer immediately begins service. Otherwise (when all servers are occupied), the customer joins the queue, which is common to all servers. The queue capacity is unlimited. When a server becomes idle, the customer at the front of the queue takes the free server. Consequently, the waiting discipline is FCFS. Service times are exponentially distributed with finite mean $1/\mu$. Times between two consecutive arrivals and service times are mutually independent.

Queueing Model M/M/s

Model description



Queueing Model M/M/s

Model Analysis

The state of the system at date t can be described using the process (5), whose state space $S = \{0, 1, 2, \dots\}$. The latter is a birth and death process whose transition rates are:

$$\lambda_n = \lambda, \forall n \geq 0 \text{ and}$$

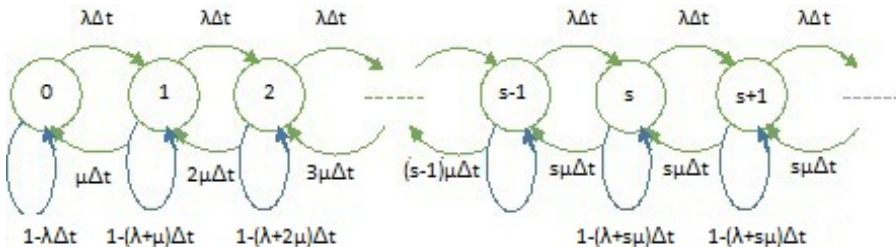
$$\mu_n = \min\{n, s\} \times \mu, n \geq 1 (\mu_0 = 0).$$

Queueing Model M/M/s

Transitory State of the system

Let the state probabilities $p_n(t) = \mathbb{P}(N(t) = n), \forall n \geq 0$.

The representative graph of transitions is



Queueing Model M/M/s

Stationary state

As before, for the limit probabilities, from the transition graph and simplification, we obtain :

$$\begin{cases} p_n = \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n p_0, 1 \leq n < s; \\ p_n = \frac{1}{s!} \frac{1}{s^{n-s}} \left(\frac{\lambda}{\mu} \right)^n p_0, n \geq s. \end{cases}$$

To calculate p_0 the probability that the system is empty we apply the normalization equation $\sum_{n=0}^{+\infty} p_n = 1$. So for $\frac{\lambda}{s\mu} < 1$ we obtain

$$p_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{\left(\frac{\lambda}{\mu} \right)^s}{s! \left(1 - \frac{\lambda}{s\mu} \right)} \right]^{-1}.$$

Queueing Model M/M/s

Stationary state

Then we get

$$p_n = \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n p_0, 1 \leq n < s$$

$$p_n = \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s \rho^{n-s} p_0, n \geq s$$

provided $\frac{\lambda}{s\mu} = \rho < 1$. ρ is the total traffic intensity.

For $n = s$ the two formulas are equal. We can write the last equation as

$$p_n = \rho^{n-s} p_s, n \geq s$$

where p_s is the probability that all servers are busy.

Queueing Model M/M/s

Performance Measures

- Mean number of customers in the system

$$\bar{n} = \frac{\lambda}{\mu} + \frac{\left(\frac{\lambda}{\mu}\right)^{s+1}}{s! \left(1 - \frac{\lambda}{s\mu}\right)^2} p_0.$$

- Mean number of customers in the queue

$$\bar{n}_q = \frac{\left(\frac{\lambda}{\mu}\right)^{s+1}}{s! \left(1 - \frac{\lambda}{s\mu}\right)^2} p_0.$$

- Mean time a customer spends in the system

$$\bar{W}_s = \frac{1}{\mu} + \frac{\left(\frac{\lambda}{\mu}\right)^s}{s\mu s! \left(1 - \frac{\lambda}{s\mu}\right)^2} p_0.$$

- Mean waiting time for a customer $\bar{W} = \frac{\left(\frac{\lambda}{\mu}\right)^s}{s! s\mu \left(1 - \rho\right)^2} p_0.$

Queueing Model $M/M/s/K$

$M/M/s/K$ Model (Limited Capacity Queueing System)

Now, let's assume that in the $M/M/s$ system, s servers are arranged in parallel, and the number of waiting positions is limited (equal to $K - s$). Upon the arrival of a customer, if all servers and waiting positions are occupied, the customer leaves the system permanently without receiving service.

Queueing Model M/M/s/K

Stochastic Analysis

The process (5) describing the state of the system is a birth and death defined on the state space $S = \{0, 1, 2, \dots, K\}$ with the transition rates

$$\begin{aligned}\lambda_n &= \lambda, 0 \leq n \leq K \text{ and} \\ \mu_n &= \begin{cases} n\mu, & \text{if } 1 \leq n \leq s-1 \\ s\mu, & \text{if } s \leq n \leq K \end{cases} \\ \mu_0 &= 0.\end{aligned}$$

Remark: Having a finite number of states, the Markov process describing the evolution of the number of customers in the queue is always ergodic, and thus the queue is stable regardless of the arrival rates λ and service rates μ .

Queueing Model M/M/s/K

Transitory State of the system

Let the state probabilities $p_n(t) = \mathbb{P}(N(t) = n), 0 \leq n \leq K$.
The system of equations for state probabilities is obtained from the graph. Then by letting $\Delta t \rightarrow 0$, we find the following Kolmogorov system of equations

$$\left\{ \begin{array}{l} p'_0(t) = -\lambda p_0(t) + \mu p_1(t) \\ \text{-----} \\ p'_n(t) = \lambda p_{n-1}(t) - (\lambda + n\mu)p_n(t) + (n+1)\mu p_{n+1}(t), \\ \qquad \qquad \qquad 1 \leq n < s \\ p'_n(t) = \lambda p_{n-1}(t) - (\lambda + s\mu)p_n(t) + s\mu p_{n+1}(t), \\ \qquad \qquad \qquad s \leq n < K \\ p'_K(t) = \lambda p_{K-1}(t) - s\mu p_K(t). \end{array} \right.$$

Queueing Model M/M/s/K

Stationary State of the System

Let the probabilities $p_n = \lim_{t \rightarrow \infty} p_n(t)$, $0 \leq n \leq K$. The stationary distribution p_n satisfies the balance equations system

$$\begin{cases} 0 = -\lambda p_0 + \mu p_1 \\ 0 = \lambda p_{n-1} - (\lambda + n\mu)p_n + (n+1)\mu p_{n+1}, & 1 \leq n < s \\ 0 = \lambda p_{n-1} - (\lambda + s\mu)p_n + s\mu p_{n+1}, & s \leq n < K \\ 0 = \lambda p_{K-1} - s\mu p_K. \end{cases}$$

Solving this system we get

$$p_n = \begin{cases} \frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n p_0, & 1 \leq n < s \\ \frac{1}{s^{n-s}s!} \left(\frac{\lambda}{\mu}\right)^n p_0 = \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s \left(\frac{\lambda}{\mu s}\right)^{n-s} p_0, & s \leq n \leq K. \end{cases}$$

Queueing Model M/M/s/K

Stationary State of the System

From $\sum_{n=0}^K p_n = 1$, the probability p_0 is obtained

$$\begin{aligned}
 p_0 &= \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s \sum_{n=s}^K \left(\frac{\lambda}{\mu s} \right)^{n-s} \right]^{-1} \\
 &= \left[\sum_{k=0}^{s-1} \frac{1}{k!} \left(\frac{\lambda}{\mu} \right)^k + \frac{\left(\frac{\lambda}{\mu} \right)^s}{s!} \frac{1 - \rho^{K-s+1}}{1 - \rho} \right]^{-1}.
 \end{aligned}$$

Queueing Model M/M/s/K

Stationary State of the System

The important measure of this system is the probability of loss, which is the probability that the system is in state K

$$p_K = \frac{1}{s!} \left(\frac{\lambda}{\mu s} \right)^{K-s} \left(\frac{\lambda}{\mu} \right)^s p_0.$$

In the particular case $K = s$ (system with blocked requests), the stationary distribution of the corresponding process is called **Erlang formula**, given by $p_n = \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n p_0, 0 \leq n \leq s$

where $p_0 = \left[\sum_{n=0}^s \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n \right]^{-1}$.

We have also that

$$P_{(loss)} = \frac{1}{s!} \left(\frac{\lambda}{\mu} \right)^s p_0.$$

Queueing Model M/M/s/K

Performance Measures

Using the stationary distribution, the mean number $\bar{n}_q$ of customers waiting in the system is

$$\begin{aligned}\bar{n}_q &= \sum_{n=1}^{K-s} n p_{s+n} \\ &= \frac{\left(\frac{\lambda}{\mu}\right)^{s+1}}{ss!} \frac{1 - \rho^{K-s} [1 + (K-s)(1-\rho)]}{(1-\rho)^2} p_0.\end{aligned}$$

and the average number $\bar{n}$ of customers present is $\bar{n} = \bar{n}_q + \frac{\lambda}{\mu}$.

Queueing Model M/M/s/K

Performance Measures

Remark: However, a difficulty arises in the system when its maximum capacity K is reached. The effective arrival rate of customers is therefore

$$\lambda_{\text{eff}} = \sum_{n=0}^{K-1} \lambda p_n = \lambda (1 - p_K).$$

Remark: The calculation of the utilization rate of each server is also based on the effective arrival rate and yields

$$U = \frac{\lambda_{\text{eff}}}{s\mu} = \rho (1 - p_K).$$

Queueing system attended by a finite number of customers (Finite source queues)

Let's assume that the population size is K and that the number of servers is less than this number K ($s < K$), otherwise there would never be any waiting.

Let's assume that the phase during which the customer does not require service is exponentially distributed with a mean of $1/\lambda$. That is, if at time t , a customer is in this phase:

- The probability that a particular source generates a service request during the interval $]t, t + \Delta t]$ is $\lambda \Delta t + o(\Delta t)$;
- The probability that a request arrives, given that there are k customers in this phase at time t is $k\lambda \Delta t + o(\Delta t)$.

Queueing system attended by a finite source queues

Model Analysis

We can use a birth and death process $\{N(t), t \geq 0\}$ to describe this type of system, whose state space is $S = \{0, 1, 2, \dots, K\}$. Note that if n is the number of customers requesting service (the number of customers already present in the system), the size of the remaining population that can generate a request is $(K - n)$. The transition rates are:

$$\lambda_n = \begin{cases} (K - n) \lambda, & 0 \leq n \leq K \\ 0, & n > K \end{cases}$$

$$\mu_n = \begin{cases} \mu \cdot \min \{n, s\}, & 1 \leq n \leq K \\ 0, & n > K \end{cases}.$$

Queueing system attended by a finite number of customers

Stationary State of the System

Let $\lim_{t \rightarrow \infty} p_n(t) = p_n$, $0 \leq n \leq K$. The stationary distribution p_n satisfies the following balance equations system

$$\left\{ \begin{array}{l} \lambda K p_0 = \mu p_1 \\ ((K - n) \lambda + n \mu) p_n = (K - n + 1) \lambda p_{n-1} + (n + 1) \mu p_{n+1}, \\ \quad 1 \leq n < s \\ ((K - n) \lambda + s \mu) p_n = (K - n + 1) \lambda p_{n-1} + s \mu p_{n+1}, \\ \quad s \leq n < K \\ s \mu p_K = \lambda p_{K-1} \end{array} \right.$$

Queueing system attended by a finite number of customers

Stationary State of the System

The solution of this system of equations is done by recursion

$$p_1 = \frac{\lambda K}{\mu} p_0$$

$$p_2 = \frac{K(K-1)}{2} \left(\frac{\lambda}{\mu}\right)^2 p_0$$

$$p_n = C_K^n \left(\frac{\lambda}{\mu}\right)^n p_0, 1 \leq n < s$$

$$p_n = C_K^n \frac{n!}{s! s^{n-s}} \left(\frac{\lambda}{\mu}\right)^n p_0, s \leq n \leq K.$$

and $p_0 = \left[\sum_{n=0}^K \frac{K!}{(K-n)!} \left(\frac{\lambda}{\mu}\right)^n \right]^{-1}.$